

Assignment 4

Coverage

Required:

Generate the functional and Code coverage reports for UART in previous assignment and it is **mandatory** to reach 100% coverage.

-Functional Coverage (100% Mandatory):

You must define **covergroups**, **coverpoints**, and **cross-coverage targets based on the specification**. Your testbench stimulus must achieve 100% Functional Coverage. No justifications are permitted for functional coverpoints.

-Code Coverage (Line, Branch, Toggle, Expression/Condition, FSM):

Your goal is 100% Code Coverage. If a code hole is reachable: You must enhance your testbench stimulus/random constraints to generate vectors that hit the uncovered code.

If a code hole is UNreachable (e.g., tie-off signals, unused states...): You must provide a Exclusion Justification explaining why the hardware cannot reach that state.

U will need to add snippets for coverage as mentioned below

Deliverables:

U have to deliver

- **Zip Folder** contains:
 - The implemented **.sv files** and the **design files** and **run.do file**.
- **Separated PDF** contains:
 - Snippets for the implemented code.
 - Snippet for the waveform shows the different testcases.
 - Snippets for the logs show the all printed values.
 - Snippets for functional coverage report.
 - Snippets for code coverage report for all parameters [Line / toggle / branch /condition/...]
 - Snippet for Do file you have used for automating the process.

The delivered zip file must be named like **your_name_Assignment_1.rar** for example:
Hassan_Khaled_Assignment_4.rar also the PDF file **Hassan_Khaled_Assignment_4.pdf**

Good Luck